

ELECTRONIC DEVICES

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC71	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To learn principle of operation, construction and characteristics of various electronic devices.
2. To study operation and characteristics of Rectifiers with filters.
3. To understand the analysis of (FET) and MOSFET characteristics.
4. To provide the concepts involved in design of electronic Circuits.

COURSE OUTCOMES:

After completing this course, the student will be able to:

1. Understand the operation and characteristics of various electronic devices.
2. Design half wave and full wave rectifiers with filters.
3. Characterize the current flow of a bipolar transistor in CB, CE and CC configurations and Biasing the transistors
4. Analyze Field effect transistor (FET) and MOSFET characteristics.
5. Design and analyze of various Oscillators.

UNIT-I	P-N JUNCTION DIODE :	
PN JUNCTION DIODE: Operation of PN junction Diode: No bias, forward bias and reverse bias, diode current equation (qualitative treatment), volt-ampere (V-I) characteristics, temperature dependence of V-I characteristics, ideal versus practical diode, static and dynamic resistances, diode equivalent circuits, breakdown mechanisms in semiconductor diodes, zener diode characteristics.		
UNIT-II	RECTIFIERS AND FILTERS :	
The P-N junction as a Rectifier, Half wave Rectifier, Full wave Rectifier, Bridge Rectifier, Harmonic components in a Rectifier Circuit, Inductor Filters, Capacitor Filters, L Section Filters, π - Section Filters, Comparison of Filters, Voltage Regulation using Zener Diode.		
UNIT-III	BIPOLAR JUNCTION TRANSISTOR AND ITS BIASING	
BIPOLAR JUNCTION TRANSISTOR (BJT): BJT construction, operation, symbol, transistor current components, input & output characteristics of a transistor in CB, CE and CC configurations TRANSISTOR BIASING AND STABILIZATION: Need for biasing, operating point, DC and AC load lines, stability factor, fixed bias circuit, collector to base bias circuit, self bias circuit.		
UNIT-IV	FIELD EFFECT TRANSISTOR	
FIELD EFFECT TRANSISTOR: Junction field effect transistor (construction, principle of operation, symbol), volt-ampere characteristics of JFET, MOSFETS (construction, principle of operation, symbol), volt- ampere characteristics of MOSFETS in enhancement and depletion modes.		
UNIT-V	OSCILLATORS	
OSCILLATORS: Barkhausen criterion, RC Phase shift oscillator using BJT, General form of LC oscillators, Hartley oscillator, Colpitts oscillator and Crystal oscillator.		

TEXT BOOKS:

1. Jacob Milliman, Christos C .Halkias, Satyabrata Jit (2015), Electronic Devices and Circuits, 3rd edition, Tata McGraw Hill, New Delhi.
2. G. K. Mittal (1999), Electronic Devices and Circuits, 22nd edition, Khanna Publications, New Delhi.
3. Robert Boylestad, Lowis Nashelsky (1993), Electronic Devices and Circuit Theory, 5th edition,

Prentice Hall of India, New Delhi, India.

REFERENCE BOOKS:

1. David. A. Bell (1986), *Electronic Devices and Circuits*, 4th edition, Prentice Hall of India, New Delhi.
2. S. Shalivahanan, N. Suresh Kumar, A. Vallavaraj (2007), *Electronic Devices and Circuits*, 3rd edition, McGraw Hill, New Delhi, India.
3. Theodore. F. Bogart Jr, Jeffrey S. Beasley, Guillermo Rico (2004), *Electronic Devices and Circuits*, 6th edition Pearson Education, India.

WEB REFERENCES:

1. <https://unacademy.com/course/electronic-devices-gate-ece/JTIAAKX1>
2. <https://freevideolectures.com/course/2261/basic-electronics-and-lab/2>
3. <https://unacademy.com/lesson/build-in-potential-and-depletion-width/ALHF5QVM>

E-TEXT BOOKS:

1. [http://www.freebookcentre.net/electronics-ebooks-download/Electronic-Devices-and-Circuits-\(PDF-313p\).html](http://www.freebookcentre.net/electronics-ebooks-download/Electronic-Devices-and-Circuits-(PDF-313p).html)
2. <https://www.goodreads.com/book/show/25345857-electronic-devices-and-circuits>
3. <https://thebookee.net/el/electronic-devices-and-circuits-by-jb-gupta-pdf>

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1. <https://www.edx.org/>
2. <https://www.coursera.org/learn/electronics>